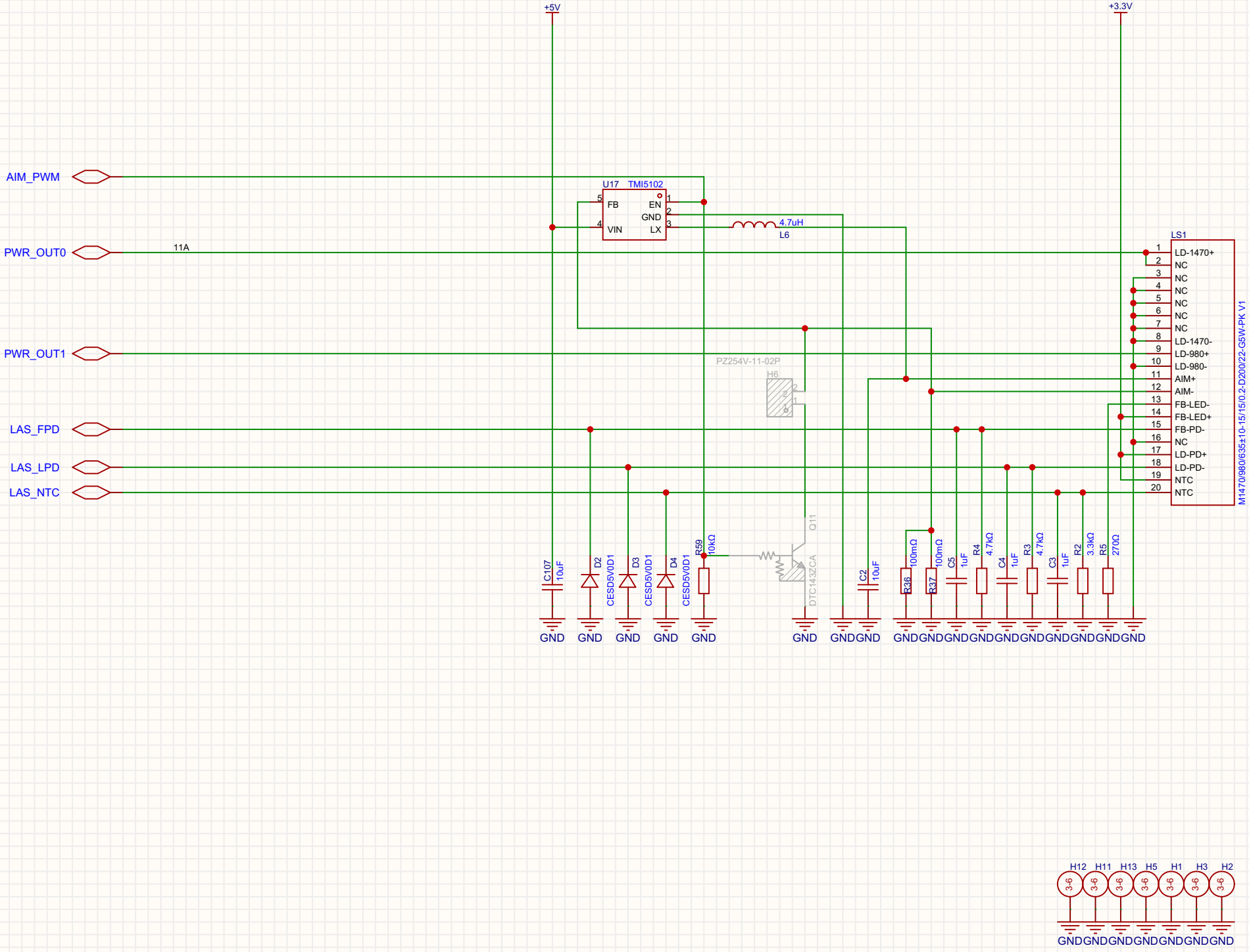
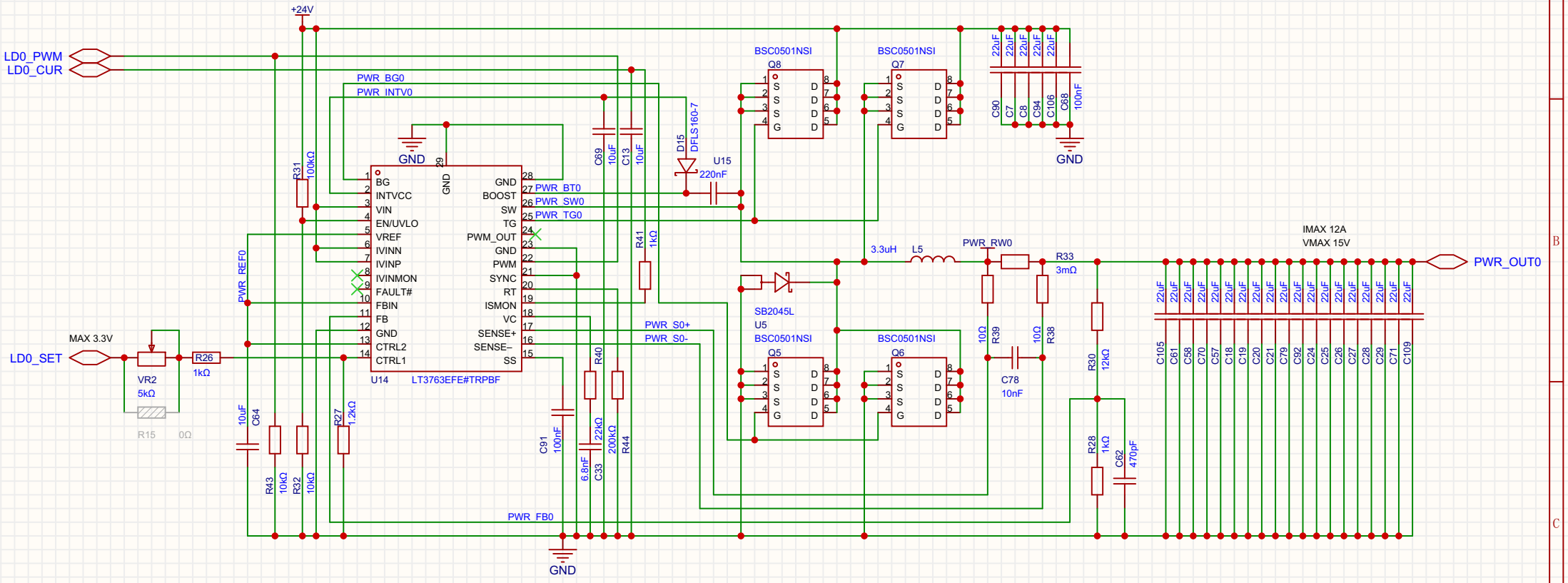
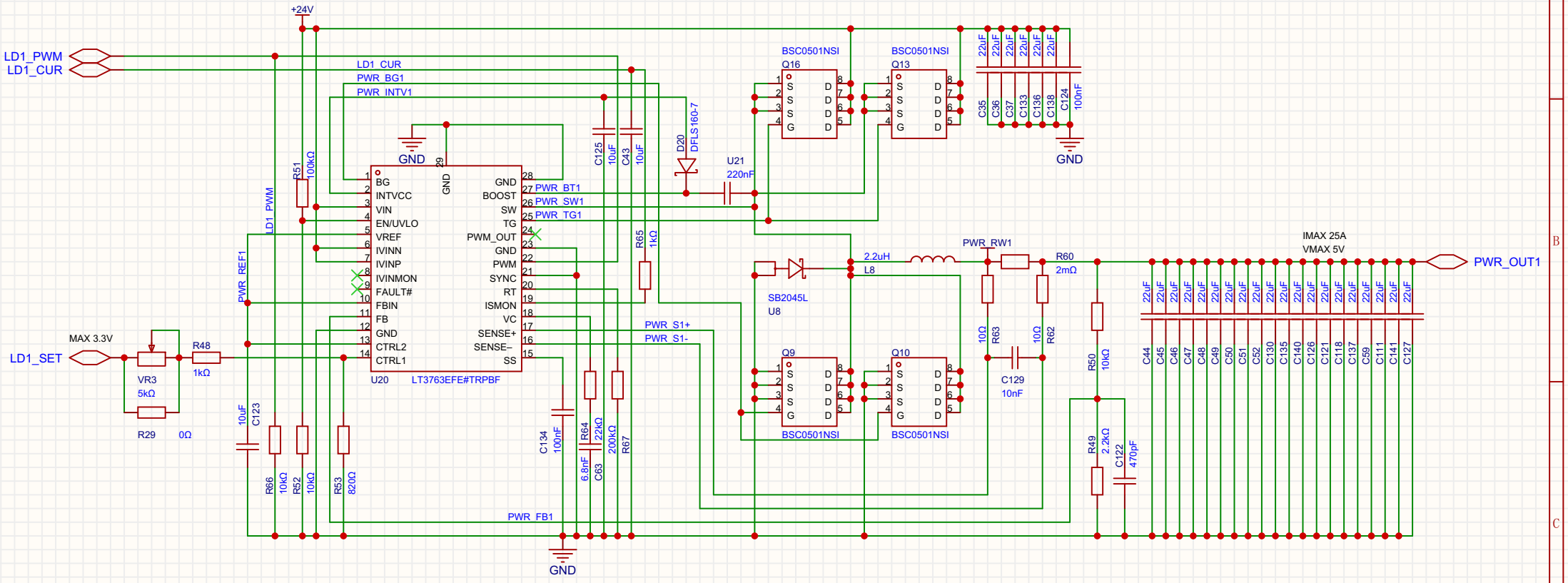
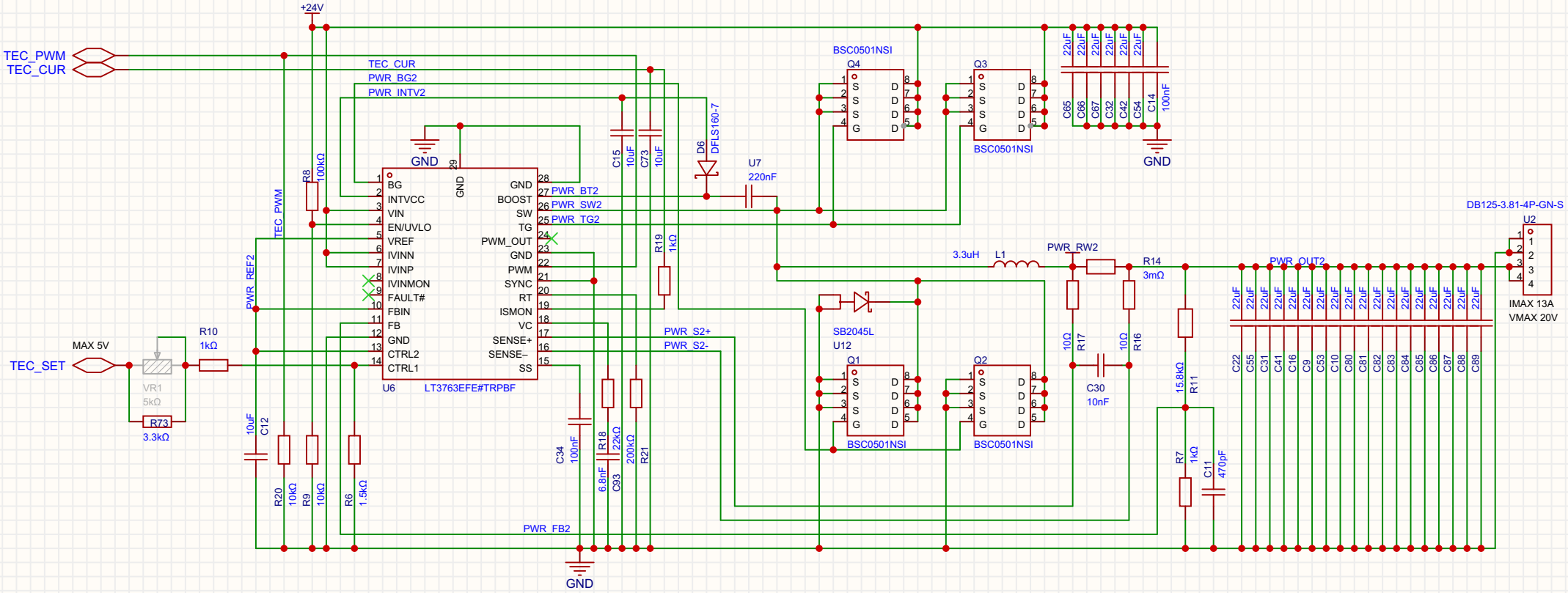
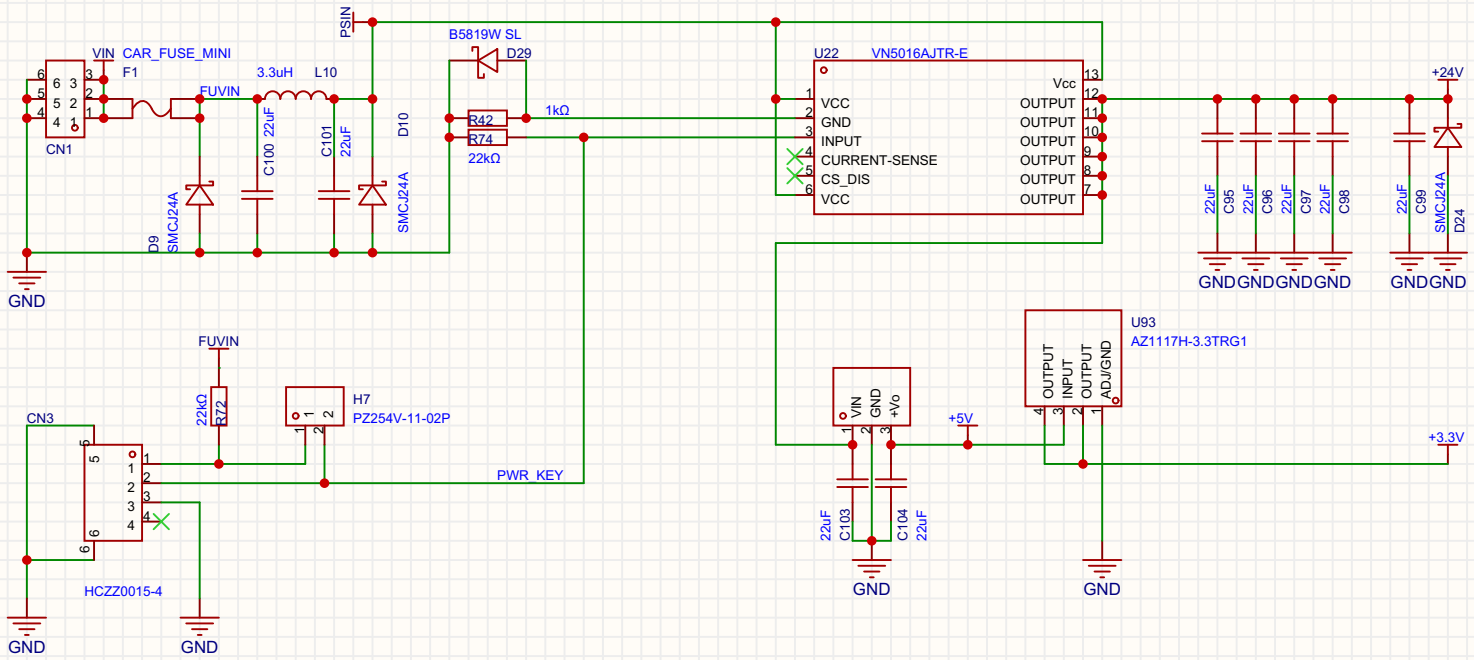
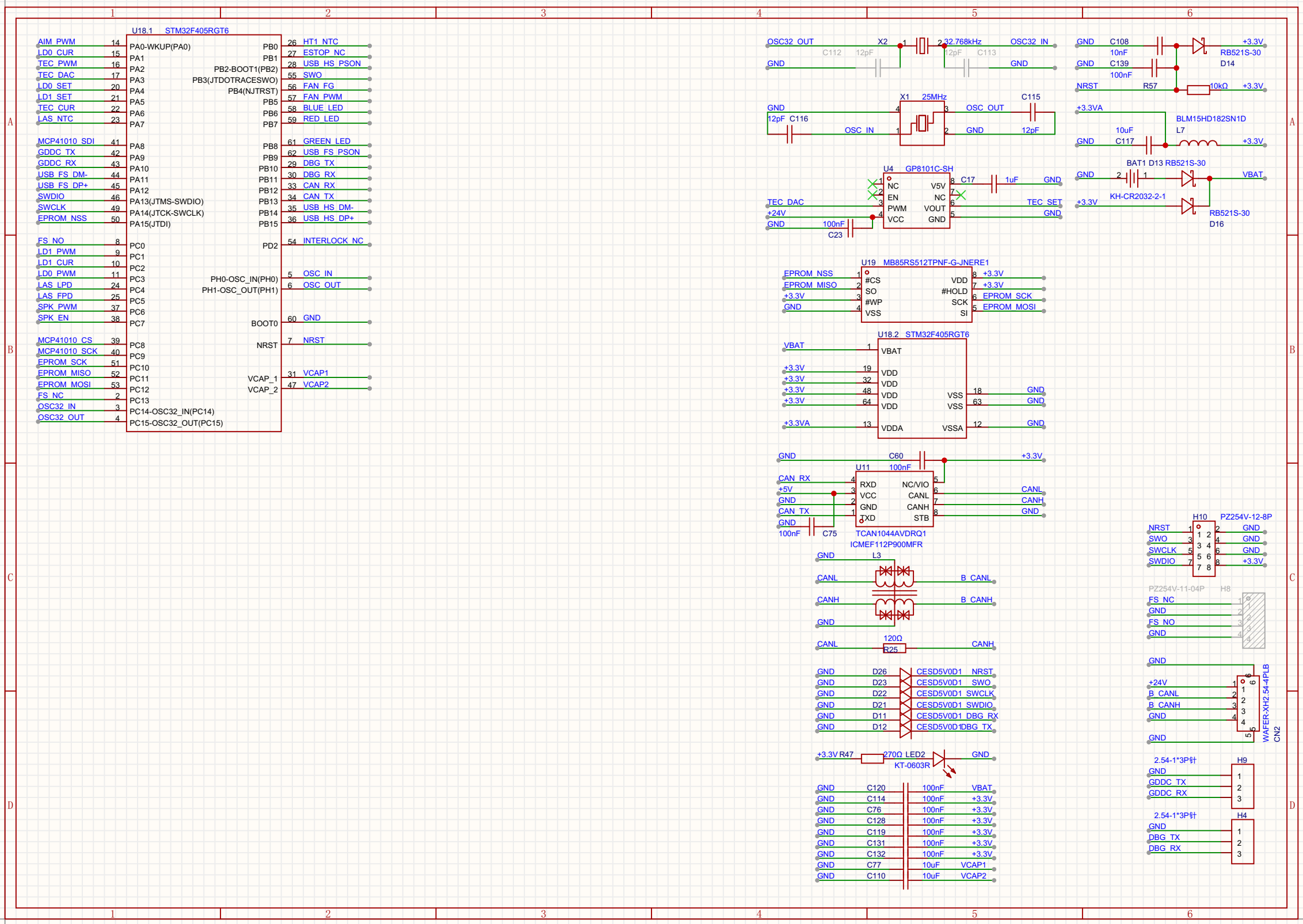

AIM_PWM
PWR_OUT0
11A
PWR_OUT1
LAS_FPD
LAS_LPD
LAS_NTC
+5V
+3.3V
U17 TMI5102
FB
VIN
EN
GND
LX
4.7uH
L6
PZ254V-11-02P
H6
LS1
LD-1470+
NC
LD-1470-
LD-980+
LD-980-
AIM+
AIM-
FB-LED-
FB-LED+
FB-PD-
LD-PD+
LD-PD-
NTC
M1470/980/635±10-15/15/0.2-D200/22-G5W-PK V1
C107 10uF
D2 CESD5V0D1
D3 CESD5V0D1
D4 CESD5V0D1
R59 10kΩ
Q11 DTC143ZCA
C2 10uF
R36 100mΩ
R37 100mΩ
C5 1uF
R4 4.7kΩ
C4 1uF
R3 4.7kΩ
C3 1uF
R2 3.3kΩ
R5 270Ω
GND
H12 H11 H13 H5 H1 H3 H2
3-6

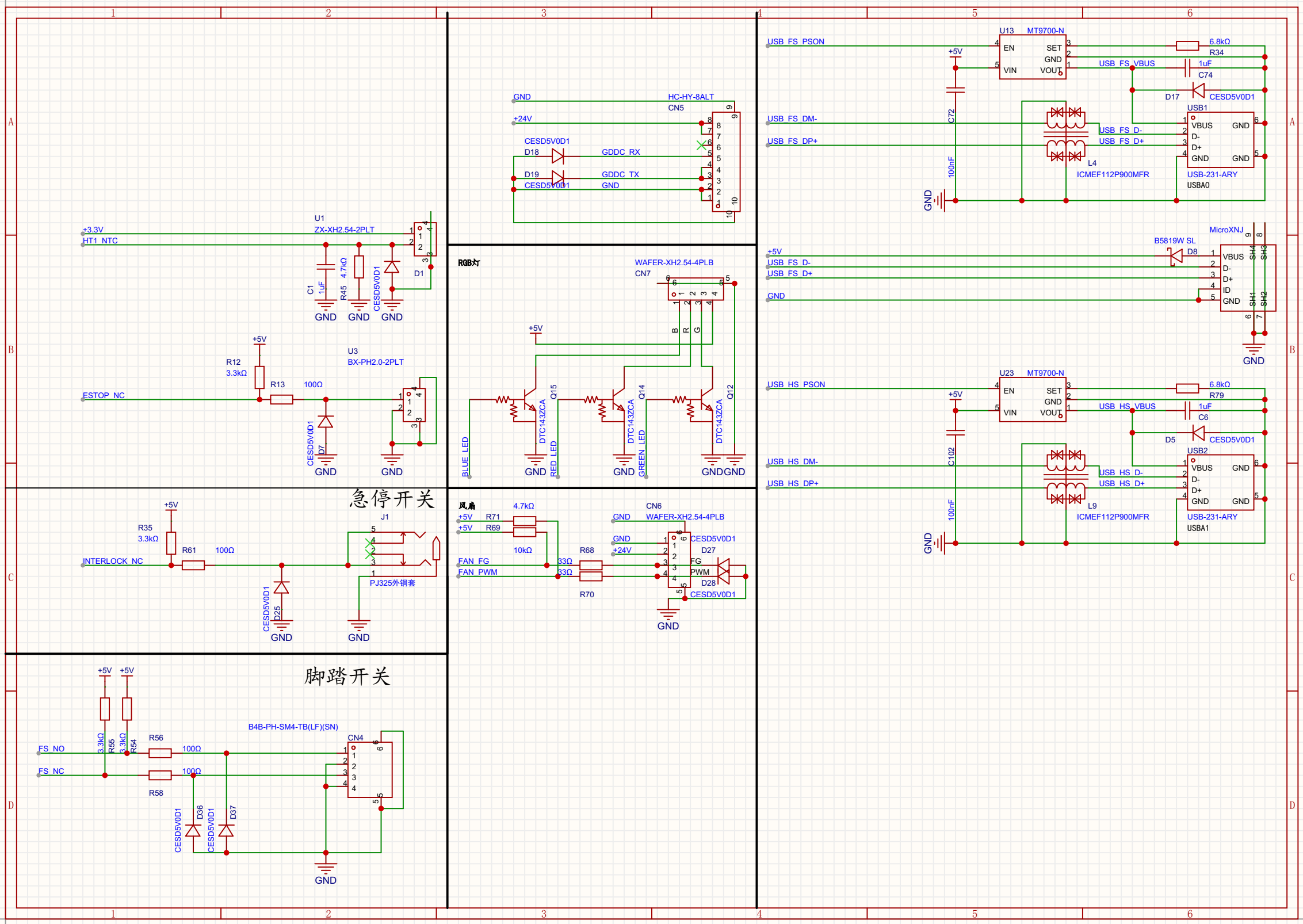

+24V
LD0_PWM
LD0_CUR
PWR_BG0
PWR_INTV0
GND
R31
100kΩ
C69
10uF
C13
10uF
D15
DFLS160-7
U15
220nF
BSC0501NSI
Q8
BSC0501NSI
Q7
22uF
C90
C7
C8
C94
C106
C68
100nF
GND
BG
INTVCC
VIN
EN/UVLO
VREF
IVINN
IVINP
IVINMON
FAULT#
FBIN
FB
GND
CTRL2
CTRL1
GND
BOOST
SW
TG
PWM_OUT
GND
PWM
SYNC
RT
ISMON
VC
SENSE+
SENSE–
SS
PWR_BT0
PWR_SW0
PWR_TG0
U14
LT3763EFE#TRPBF
R41
1kΩ
PWR_REF0
3.3uH
L5
PWR_RW0
R33
3mΩ
IMAX 12A
VMAX 15V
PWR_OUT0
SB2045L
U5
BSC0501NSI
Q5
BSC0501NSI
Q6
10Ω
R39
10Ω
R38
C78
10nF
MAX 3.3V
LD0_SET
R26
1kΩ
VR2
5kΩ
R15
0Ω
10uF
C64
R27
1.2kΩ
R40
22kΩ
200kΩ
R44
C91
100nF
6.8nF
C33
PWR_S0+
PWR_S0-
R30
12kΩ
R28
1kΩ
C62
470pF
R43
10kΩ
R32
10kΩ
PWR_FB0
GND
C105
C61
C58
C70
C57
C18
C19
C20
C21
C79
C92
C24
C25
C26
C27
C28
C29
C71
C109

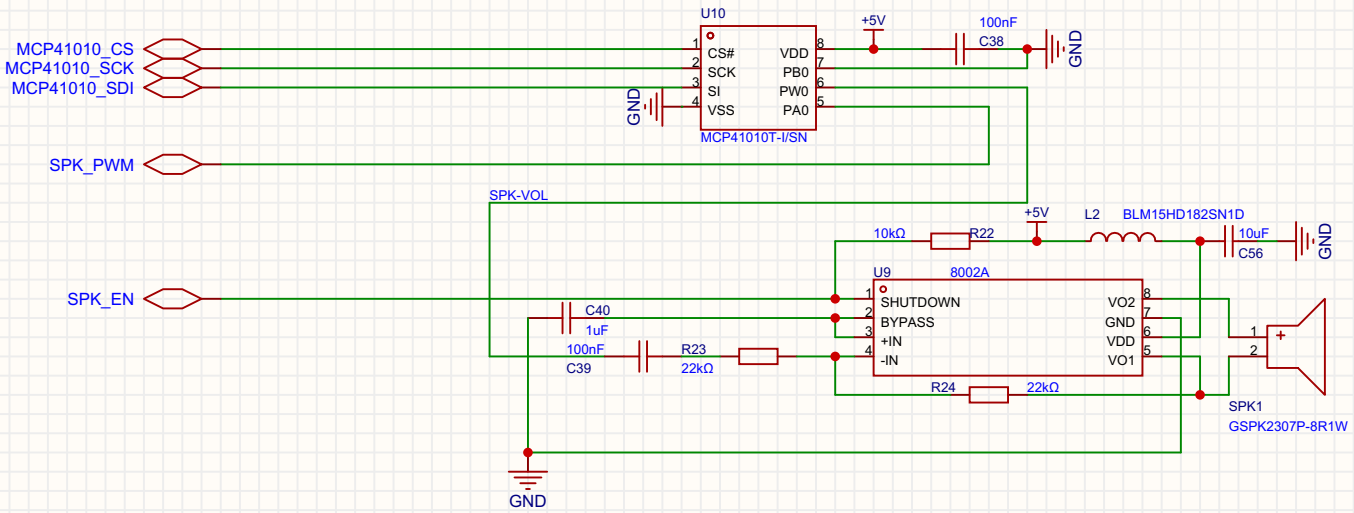

+24V
LD1_PWM
LD1_CUR
PWR_BG1
PWR_INTV1
PWR_REF1
LD1_SET
MAX 3.3V
VR3
5kΩ
R29
0Ω
R48
1kΩ
R51
100kΩ
C123
10uF
R66
10kΩ
R52
10kΩ
R53
820Ω
U20
LT3763EFE#TRPBF
BG
INTVCC
VIN
EN/UVLO
VREF
IVINN
IVINP
IVINMON
FAULT#
FBIN
FB
GND
CTRL2
CTRL1
GND
BOOST
SW
TG
PWM_OUT
GND
PWM
SYNC
RT
ISMON
VC
SENSE+
SENSE–
SS
PWR_BT1
PWR_SW1
PWR_TG1
C125
10uF
C43
10uF
D20
DFLS160-7
U21
220nF
R65
1kΩ
C134
100nF
R64
22kΩ
C63
6.8nF
R67
200kΩ
PWR_S1+
PWR_S1-
PWR_FB1
BSC0501NSI
Q16
Q13
Q9
Q10
SB2045L
U8
2.2uH
L8
PWR_RW1
R60
2mΩ
R63
10Ω
R62
10Ω
C129
10nF
R50
10kΩ
R49
2.2kΩ
C122
470pF
22uF
C35
C36
C37
C133
C136
C138
C124
100nF
C44
C45
C46
C47
C48
C49
C50
C51
C52
C130
C135
C140
C126
C121
C118
C137
C59
C111
C141
C127
IMAX 25A
VMAX 5V
PWR_OUT1
GND


+24V
TEC_PWM
TEC_CUR
TEC_SET
MAX 5V
VR1
5kΩ
R73
3.3kΩ
R10
1kΩ
PWR_REF2
TEC_CUR
PWR_BG2
PWR_INTV2
TEC_PWM
R8
100kΩ
GND
U6
LT3763EFE#TRPBF
BG
INTVCC
VIN
EN/UVLO
VREF
IVINN
IVINP
IVINMON
FAULT#
FBIN
FB
GND
CTRL2
CTRL1
GND
BOOST
SW
TG
PWM_OUT
GND
PWM
SYNC
RT
ISMON
VC
SENSE+
SENSE–
SS
PWR_BT2
PWR_SW2
PWR_TG2
PWR_S2+
PWR_S2-
PWR_FB2
C12
10uF
R20
10kΩ
R9
10kΩ
R6
1.5kΩ
C34
100nF
R18
22kΩ
C93
6.8nF
R21
200kΩ
C15
10uF
C73
10uF
R19
1kΩ
D6
DFLS160-7
U7
220nF
BSC0501NSI
Q4
Q3
Q1
Q2
SB2045L
U12
3.3uH
L1
PWR_RW2
R14
3mΩ
R17
10Ω
R16
10Ω
C30
10nF
R11
15.8kΩ
R7
1kΩ
C11
470pF
22uF
C65
C66
C67
C32
C42
C54
C14
100nF
PWR_OUT2
C22
C55
C31
C41
C16
C9
C53
C10
C80
C81
C82
C83
C84
C85
C86
C87
C88
C89
DB125-3.81-4P-GN-S
U2
IMAX 13A
VMAX 20V


PSIN
VIN
CAR_FUSE_MINI
F1
CN1
FUVIN
3.3uH
L10
C100
22uF
C101
22uF
D10
SMCJ24A
D9
SMCJ24A
B5819W SL
D29
R42
R74
22kΩ
1kΩ
U22
VN5016AJTR-E
VCC
GND
INPUT
CURRENT-SENSE
CS_DIS
VCC
Vcc
OUTPUT
OUTPUT
OUTPUT
OUTPUT
OUTPUT
OUTPUT
+24V
22uF
C95
22uF
C96
22uF
C97
22uF
C98
22uF
C99
SMCJ24A
D24
GND
GND
GND
GND
GND
GND
FUVIN
22kΩ
R72
H7
PZ254V-11-02P
CN3
HCZZ0015-4
PWR_KEY
GND
GND
U93
AZ1117H-3.3TRG1
VIN
GND
+Vo
OUTPUT
INPUT
OUTPUT
ADJ/GND
+5V
+3.3V
22uF
C103
C104
22uF
GND
GND


U18.1 STM32F405RGT6
AIM_PWM 14 PA0-WKUP(PA0)
LD0_CUR 15 PA1
TEC_PWM 16 PA2
TEC_DAC 17 PA3
LD0_SET 20 PA4
LD1_SET 21 PA5
TEC_CUR 22 PA6
LAS_NTC 23 PA7
MCP41010_SDI 41 PA8
GDDC_TX 42 PA9
GDDC_RX 43 PA10
USB_FS_DM- 44 PA11
USB_FS_DP+ 45 PA12
SWDIO 46 PA13(JTMS-SWDIO)
SWCLK 49 PA14(JTCK-SWCLK)
EPROM_NSS 50 PA15(JTDI)
FS_NO 8 PC0
LD1_PWM 9 PC1
LD1_CUR 10 PC2
LD0_PWM 11 PC3
LAS_LPD 24 PC4
LAS_FPD 25 PC5
SPK_PWM 37 PC6
SPK_EN 38 PC7
MCP41010_CS 39 PC8
MCP41010_SCK 40 PC9
EPROM_SCK 51 PC10
EPROM_MISO 52 PC11
EPROM_MOSI 53 PC12
FS_NC 2 PC13
OSC32_IN 3 PC14-OSC32_IN(PC14)
OSC32_OUT 4 PC15-OSC32_OUT(PC15)
PB0 26 HT1_NTC
PB1 27 ESTOP_NC
PB2-BOOT1(PB2) 28 USB_HS_PSON
PB3(JTDOTRACESWO) 55 SWO
PB4(NJTRST) 56 FAN_FG
PB5 57 FAN_PWM
PB6 58 BLUE_LED
PB7 59 RED_LED
PB8 61 GREEN_LED
PB9 62 USB_FS_PSON
PB10 29 DBG_TX
PB11 30 DBG_RX
PB12 33 CAN_RX
PB13 34 CAN_TX
PB14 35 USB_HS_DM-
PB15 36 USB_HS_DP+
PD2 54 INTERLOCK_NC
PH0-OSC_IN(PH0) 5 OSC_IN
PH1-OSC_OUT(PH1) 6 OSC_OUT
BOOT0 60 GND
NRST 7 NRST
VCAP_1 31 VCAP1
VCAP_2 47 VCAP2
OSC32_OUT X2 1 2 32.768kHz OSC32_IN
C112 12pF 12pF C113
GND GND
X1 25MHz C115
GND 4 3 OSC_OUT
12pF C116 OSC_IN 1 2 GND 12pF
U4 GP8101C-SH
1 NC V5V 8 C17 1uF GND
2 EN NC 7
TEC_DAC 3 PWM VOUT 6 TEC_SET
+24V 4 VCC GND 5 GND
GND 100nF C23
U19 MB85RS512TPNF-G-JNERE1
EPROM_NSS 1 #CS VDD 8 +3.3V
EPROM_MISO 2 SO #HOLD 7 +3.3V
+3.3V 3 #WP SCK 6 EPROM_SCK
GND 4 VSS SI 5 EPROM_MOSI
U18.2 STM32F405RGT6
VBAT 1 VBAT
+3.3V 19 VDD
+3.3V 32 VDD
+3.3V 48 VDD VSS 18 GND
+3.3V 64 VDD VSS 63 GND
+3.3VA 13 VDDA VSSA 12 GND
GND C60 +3.3V
U11 100nF
CAN_RX 4 RXD NC/VIO 5
+5V 3 VCC CANL 6 CANL
GND 2 GND CANH 7 CANH
CAN_TX 1 TXD STB 8 GND
GND 100nF C75 TCAN1044AVDRQ1
ICMEF112P900MFR
GND L3
CANL B_CANL
CANH B_CANH
GND
CANL 120Ω CANH
R25
GND D26 CESD5V0D1 NRST
GND D23 CESD5V0D1 SWO
GND D22 CESD5V0D1 SWCLK
GND D21 CESD5V0D1 SWDIO
GND D11 CESD5V0D1 DBG_RX
GND D12 CESD5V0D1 DBG_TX
+3.3V R47 270Ω LED2 GND
KT-0603R
GND C120 100nF VBAT
GND C114 100nF +3.3V
GND C76 100nF +3.3V
GND C128 100nF +3.3V
GND C119 100nF +3.3V
GND C131 100nF +3.3V
GND C132 100nF +3.3V
GND C77 10uF VCAP1
GND C110 10uF VCAP2
GND C108 10nF +3.3V
GND C139 100nF RB521S-30 D14
NRST R57 10kΩ +3.3V
+3.3VA BLM15HD182SN1D
10uF L7
GND C117 +3.3V
BAT1 D13 RB521S-30
GND 2 1 VBAT
KH-CR2032-2-1
+3.3V RB521S-30 D16
H10 PZ254V-12-8P
NRST 1 2 GND
SWO 3 4 GND
SWCLK 5 6 GND
SWDIO 7 8 +3.3V
PZ254V-11-04P H8
FS_NC 1
GND 2
FS_NO 3
GND 4
GND
+24V 1
B_CANL 2
B_CANH 3
GND 4
GND 5 6
WAFER-XH2.54-4PLB CN2
2.54-1*3P针 H9
GND 1
GDDC_TX 2
GDDC_RX 3
2.54-1*3P针 H4
GND 1
DBG_TX 2
DBG_RX 3


U1
ZX-XH2.54-2PLT
+3.3V
HT1_NTC
C1
1uF
R45
4.7kΩ
CESD5V0D1
D1
GND
U3
BX-PH2.0-2PLT
+5V
R12
3.3kΩ
R13
100Ω
ESTOP_NC
急停开关
J1
R35
3.3kΩ
R61
100Ω
INTERLOCK_NC
PJ325外铜套
CESD5V0D1
D25
脚踏开关
B4B-PH-SM4-TB(LF)(SN)
CN4
FS_NO
FS_NC
R55
3.3kΩ
R54
3.3kΩ
R56
100Ω
R58
100Ω
D36
D37
GND
HC-HY-8ALT
CN5
+24V
CESD5V0D1
D18
GDDC_RX
D19
GDDC_TX
RGB灯
WAFER-XH2.54-4PLB
CN7
B
R
G
DTC1432CA
Q15
Q14
Q12
BLUE_LED
RED_LED
GREEN_LED
风扇
R71
4.7kΩ
R69
10kΩ
FAN_FG
FAN_PWM
R68
33Ω
R70
33Ω
CN6
WAFER-XH2.54-4PLB
D27
D28
FG
PWM
U13
MT9700-N
EN
SET
GND
VIN
VOUT
USB_FS_PSON
USB_FS_DM-
USB_FS_DP+
USB_FS_VBUS
R34
6.8kΩ
C74
1uF
D17
C72
100nF
L4
ICMEF112P900MFR
USB1
USB-231-ARY
USBA0
USB_FS_D-
USB_FS_D+
MicroXNJ
B5819W SL
D8
VBUS
ID
SH1
SH2
SH3
SH4
U23
MT9700-N
USB_HS_PSON
USB_HS_DM-
USB_HS_DP+
USB_HS_VBUS
R79
6.8kΩ
C6
1uF
D5
C102
100nF
L9
ICMEF112P900MFR
USB2
USB-231-ARY
USBA1
USB_HS_D-
USB_HS_D+


MCP41010_CS
MCP41010_SCK
MCP41010_SDI
SPK_PWM
SPK_EN
SPK-VOL
U10
CS#
SCK
SI
VSS
VDD
PB0
PW0
PA0
MCP41010T-I/SN
GND
+5V
100nF
C38
U9
8002A
SHUTDOWN
BYPASS
+IN
-IN
VO2
GND
VDD
VO1
10kΩ
R22
L2
BLM15HD182SN1D
10uF
C56
C40
1uF
100nF
C39
R23
22kΩ
R24
22kΩ
SPK1
GSPK2307P-8R1W